

Line 12-19: Iterates through each record in the database, prepares HTML for check-box and task details.

Line 12: `localStorage.length` returns the number of records present in the database.

Line 14: Fetches the unique key using the function `localStorage.key()`.

Line 15: Fetches the complete record—fields separated by semi-colon, using the unique key.

Line 16: Extracts fields, separating them using `split()`.

Line 17: Creates an HTML form element for each task (check-box followed by task details). Inserts the function `deleteTask()` as part of the actions when the user completes a task and ticks the check-box. To delete a task from the database, we need the unique key—which is the value of each check-box.

Line 18: Appends tasks, and creates one HTML string that will be output below the New Task form, replacing the previous task list with the new one using the `innerHTML` attribute.

```
1: function deleteTask(key)
2: {
3:   localStorage.removeItem(key);
4:   getTasks();
5: }
```

Line 3: When the user clicks the check-box to mark it as complete, the record key is passed to `deleteTask()`, which uses `localStorage.removeItem()` to remove the record.

Line 4: The list of pending tasks is refreshed after the deletion.

Extending MyLocalTodo to a full application

Here are some ideas that could be implemented to build on the skeleton app:

- Add colour to tasks: Differentiate tasks with different priorities with colours. Tasks past the due date can be in RED.
- zMake the due date ‘human-friendly’: Add flexibility to accept human-friendly due dates like ‘today’, ‘tomorrow’, ‘by next week’, ‘end of March’, etc.
- Export/import tasks: Due to localStorage limitations, the task database is restricted to one browser. Extend the app

to export records as a CSV file to make tasks portable across browsers, and for back-up purposes.

- Import CSV: The ability to import a CSV file into localStorage could allow the import of tasks from Outlook, Google Tasks, etc.
- Integration with Phonebook: A more friendly to-do app could integrate with the Phonebook application. Tasks that involve communication (a phone call or email) can fetch the phone number/email address and add it to the task, to enable calling or emailing the person. For example: for a task like ‘call @ Rohit on Sunday’, it displays ‘call Rohit (9384948392) on Sunday’, so that you can dial by clicking the number.
- Add tags to group tasks: Each task can have a tag such as ‘personal’, ‘work’, ‘call’ or ‘email’, to group the related tasks to be performed together.

Develop other applications using localStorage

Some quick ideas:

- Phonebook: The MyLocalTodo example app can be modified to develop a Phonebook application.
- Project Time Tracker: You can find the `monkeeTime` application, which is a Project Time Tracker, at <https://github.com/monkeecreate/monkeeTime>. 

References

- [1] My first introduction to HTML localStorage was through the article <http://paperkitltdrock.com/2010/05/html5-localstorage-part-one/>
- [2] A good introduction to HTML5: <http://diveintohtml5.info/storage.html>
- [3] Standard reference: <http://dev.w3.org/html5/webstorage>
- [4] The source code in this article can be obtained from http://www.linuxforu.com/article_source_code/apr12/local_storage.zip

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